SOURI SASANFAR

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RESEARCH INTERESTS

Human–AI interaction and decision making; experiments and evaluation of interactive and AI systems; natural language processing and large language models for health; simulation and validation of human behavior; sequential and latent-state models of users; responsible AI.

EDUCATION

Ph.D., Industrial Engineering, Northeastern University, Boston, MA *Expected May 2027*

- GPA 3.9/4.0 (Rank 1st in cohort); ranked 1st in the Ph.D. candidacy examination.
- Dissertation: *Context-Aware Modeling for High-Stakes Decision Support in Healthcare and Pharmaceutical Supply Chains*
- Advisor: Prof. Jacqueline Griffin

M.Sc., Industrial Engineering, Iran University of Science and Technology (IUST), Tehran, Iran *Jan 2016*

- GPA 4.0/4.0 (Rank 1st).

B.Sc., Mechanical Engineering, Semnan University, Iran *Sep 2012*

PUBLICATIONS

Peer-Reviewed Journal Articles

1. Taseh A, **Sasanfar S**, Chan M, Sirls E, Nazarian A, Batmanghelich K, Bean JF, Ashkani-Esfahani S. Performance of Natural Language Processing versus International Classification of Diseases Codes in Building Registries for Patients With Fall Injury: Retrospective Analysis. *JMIR Medical Informatics*. 2025;13:e66973. https://doi.org/10.2196/66973
2. **Sasanfar S**, Bagherpour M, Moatari-Kazerouni A. Improving Emergency Departments: Simulation-Based Optimization of Patients' Waiting Time and Staff Allocation in an Iranian Hospital. *International Journal of Healthcare Management*. 2021;14(4):1449–1456. https://doi.org/10.1080/20479700.2020.1765121

Peer-Reviewed Conference Proceedings

3. **Sasanfar S**, Mohaddesi O, Chicoine N, Griffin J, Harteveld C. Validating Agents Through Simulated and Behavioral Data for Pharmaceutical Supply Chains. *Proceedings of the 2025 Winter Simulation Conference (WSC)*. 2025.

Working Papers

4. **Sasanfar S**, Mohaddesi O, Lainfiesta X, Griffin J, Harteveld C, Ergun O. The Role of Upstream Information Sharing on Human Behavior in Shortage-Driven Pharmaceutical Supply Chains. **Job market paper.** Under review, *Production and Operations Management*. 2026.
5. Taseh A, **Sasanfar S**, Chan M, Sirls E, Nazarian A, Batmanghelich K, Bean JF, Ashkani-Esfahani S. Risk Factors for Recurrent Falls in Patients with Type 2 Diabetes Following Emergency Department Fall-Related Visits. Submitted, *Archives of Gerontology and Geriatrics*. 2026.
6. **Sasanfar S**, Gerdesmeyer L, Ashkani-Esfahani S, Taseh A. Context-Aware NLP Post-Processing Improves Machine Learning Fall Detection in T2DM Emergency Department Documentation. In preparation; to be submitted to *JMIR Medical Informatics*.

CONFERENCE PRESENTATIONS

- 2026 INFORMS Annual Meeting, San Francisco, CA. Session "AI Analytics for Healthcare Safety," November 2, 2026.
- Validating Agents Through Simulated and Behavioral Data for Pharmaceutical Supply Chains. 2025 Winter Simulation Conference.

GRANTS & RESEARCH SUPPORT

- Graduate Research Assistant and key contributor — NSF Award No. 2228510, "Designing an Improved Information Infrastructure for Better Decision Making in Pharmaceutical Supply Chains" ($750,000; PI: J. Griffin; NSF Strengthening American Infrastructure program). Developed new models of human decision-making under varying information availability and co-led the design of decision-support system components.

RESEARCH EXPERIENCE

Graduate Research Assistant, Northeastern University *Sep 2022 – Present*

- Built behavioral and agent-based simulations of wholesaler decision-making under pharmaceutical supply-chain disruption; applied PCA, clustering, and Hidden Markov Models to classify ordering archetypes from human-in-the-loop experimental game data.
- Contributed to the design and analysis of a controlled between-subjects experiment (178 participants) on upstream information sharing and defensive ordering under drug shortages.

Co-Researcher, FARIL Lab, Mass General Brigham / Harvard Medical School *Jan 2024 – Present*

- Designed hybrid clinical NLP pipelines for automated fall detection and fall-mechanism classification from emergency-department notes.
- Layered deterministic, context-aware rule-based post-processing (medspaCy / ConText-style negation, temporality, and risk-vs-event rules) on AdaBoost and XGBoost classifiers across 26,975 ED notes from 670 patients with type 2 diabetes, substantially reducing false positives while preserving sensitivity.

Independent Research Project, Northeastern University

- Multi-country sentiment and semantic analysis of menstrual-stigma discourse on social media (NL, IT, TR, IR), linking sentiment to gender-gap indices via LIWC- and Python-based NLP and social-network analysis.

TECHNICAL SKILLS & METHODS

- **Programming:** Python, R, SQL; Git/GitHub, REST APIs, FastAPI.
- **Machine Learning & NLP:** scikit-learn, XGBoost, AdaBoost; spaCy/medspaCy, ConText/NegEx-style context-aware rules; classification, clustering, PCA, Hidden Markov Models.
- **Deep Learning:** PyTorch, TensorFlow.
- **LLM Engineering & RAG:** prompt engineering, structured/JSON output, tool/function calling, embeddings and vector databases (FAISS, Chroma); OpenAI and Anthropic APIs, LangChain.
- **Simulation & OR:** Arena (discrete-event simulation), OptQuest optimization, agent-based modeling.
- **Data Tooling:** dbt, Snowflake, Tableau, SPSS.

PROFESSIONAL & INDUSTRY EXPERIENCE

Data Scientist Intern, Apploi, Remote / Boston, MA *Jun 2025 – Sep 2025*

- Built an LLM-powered pipeline (few-shot prompting, JSON-schema-enforced structured output) to normalize applicant-tracking statuses and perform sentiment analysis on recruiter notes; integrated outputs into a downstream ML ranking model with explicit output validation.
- Developed a supervised ML candidate-ranking model in Snowflake/dbt (Python, scikit-learn); shipped as reproducible, code-reviewed dbt models.

Systems and Processes Analyst, Hermes Institute of Science and Technology, Tehran, Iran *Jul 2019 – Dec 2021*

- Identified workflow inefficiencies and improved overall system efficiency by 27%; built internal Python tooling and led requirement analysis for a managerial dashboard tracking 30+ KPIs.

TEACHING EXPERIENCE

Guest Lecturer — Northeastern University

- IE 6205: Deterministic Operations Research (Fall 2026).
- Simulation Modeling and Analysis (Spring 2023).
- Simulation Analysis Workshop — invited lecturer (Jan 2024).

Graduate Teaching Assistant — Northeastern University (Sep 2022 – Present)

- IE 7280: Statistical Methods in Engineering — co-designed and delivered course materials and led multiple class sessions (responsibilities beyond a standard TA role).
- IE 6205: Deterministic Operations Research.
- IE 4515: Operations Research.
- Simulation Modeling and Analysis — taught Arena and Python for applied analytics.
- Engineering Database Systems; Engineering Project Management.
- Mentored 180+ undergraduate and graduate students on technical topics, research methods, and career development.

- Developed a structured training repository of learning materials and technical resources for incoming Ph.D. students.

HONORS & AWARDS

- Yamamura Award — Teaching PhD Award, Dept. of Mechanical & Industrial Engineering, Northeastern University (2025).
- Ranked 1st, Ph.D. Candidacy Examination, Northeastern University (2023).
- 1st GPA in Industrial Engineering, two consecutive semesters, Northeastern University (2022–2023).
- Top Student Award, Iran University of Science and Technology (2015).

PROFESSIONAL SERVICE

- Peer Reviewer (2024–2026): Journal of the Operational Research Society; Decision Analytics Journal; International Journal of Healthcare Management; Frontiers in Public Health; Academy of Management Annual Meeting (Health Care Management Division).

CERTIFICATIONS

- LEADERs Leadership Certificate Program, Gordon Institute of Engineering Leadership and the PhD Network, Northeastern University (2026).
- Python for Data Science and Machine Learning; NLP Specialization; Generative AI Applications Specialist.

REFERENCES

- Prof. Jacqueline Griffin — Dept. of Mechanical & Industrial Engineering, Northeastern University (ja.griffin@northeastern.edu)
- Prof. Ozlem Ergun — Dept. of Mechanical & Industrial Engineering, Northeastern University (o.ergun@northeastern.edu)
- Prof. Casper Harteveld — College of Arts, Media & Design, Northeastern University (c.harteveld@northeastern.edu)